MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION SYSTEMS.

Programming Concepts

Discussed questions and their outputs.

1. Write a C program to read student registration number and marks of three subjects and calculate the total, percentage and division.

```c
#include<stdio.h>
main()
{
float reg,m1,m2,m3;
printf("Enter your registration no: \n");
scanf("%f",&reg);
printf("Enter three subject marks:\n");

printf(" Enter the first marks:  ");
scanf("%f",&m1);
printf(" Enter the second marks: ");
scanf("%f",&m2);
printf(" Enter the third marks:  ");
scanf("%f",&m3);

printf("The total is: %f \n",m1+m2+m3);
printf("The average is: %f \n",(m1+m2+m3)/3);
printf("The percentage is: %f \n",(m1+m2+m3)/300*100);
}
```

```
Quincy 2005
Enter your registration no:
22100934340037
Enter three subject marks:
 Enter the first marks:  70
 Enter the second marks: 87
 Enter the third marks:  76
The total is: 233.000000
The average is: 77.666667
The percentage is: 77.666667

Press Enter to return to Quincy...
```

2. Write a program that computes the area of a circle, the area of a rectangle and the area of a triangle. Let the program prompt the user to enter dimensions, perform calculations and print the result on each case.

```c
#include<stdio.h>
main()
{

float R,L,W,B,H,A;
 {
 printf("The program allows you to enter the dimensins; \n\n");
 printf("Enter the radius of the circle: \n");
 scanf("%f",&R);
 A = 3.14*R*R;
 printf("The area of the circle is %f \n\n\n",A);
 }

{
 printf("Enter the length of the rectangle: \n");
 scanf("%f",&L);
 printf("Enter the width of the rectangle: \n");
 scanf("%f",&W);
 A = L*W;
 printf("The area of the rectangle is %f\n\n\n",A);
}

{
printf("Enter the base of the tringle: \n");
scanf("%f",&B);
printf("Enter the height of the triangle: \n");
scanf("%f",&H);
A = 0.5*B*H;
printf("The area of triangle is %f",A);
}

}
```

```
Quincy 2005
The program allows you to enter the dimensins;

Enter the radius of the circle:
14
The area of the circle is 615.440002


Enter the length of the rectangle:
12
Enter the width of the rectangle:
2
The area of the rectangle is 24.000000


Enter the base of the tringle:
10
Enter the height of the triangle:
12
The area of triangle is 60.000000
Press Enter to return to Quincy...
```

3. Write a C program that reads ID number, sales price and sales quantity of an item and compute sales amount. The display results should be in the following format;

   The sales amount for an Item No. xxxx is yyyy

```c
#include<stdio.h>
main()
{
int ID;
float price,quantity,amount;
printf("enter ID number: ");
scanf("%d",&ID);
printf("enter sales price: ");
scanf("%f",&price);
printf("enter sales quantity: ");
scanf("%f",&quantity);
amount = quantity*price;
printf(" The sales amount for an item No:%d is %f",ID,amount);

}
```

```
Quincy 2005
enter ID number: 37
enter sales price: 200
enter sales quantity: 20
 The sales amount for an item No:37 is 4000.000000
Press Enter to return to Quincy...
```

4. Write a C program to calculate simple interest of a given amount for a year entered by user.

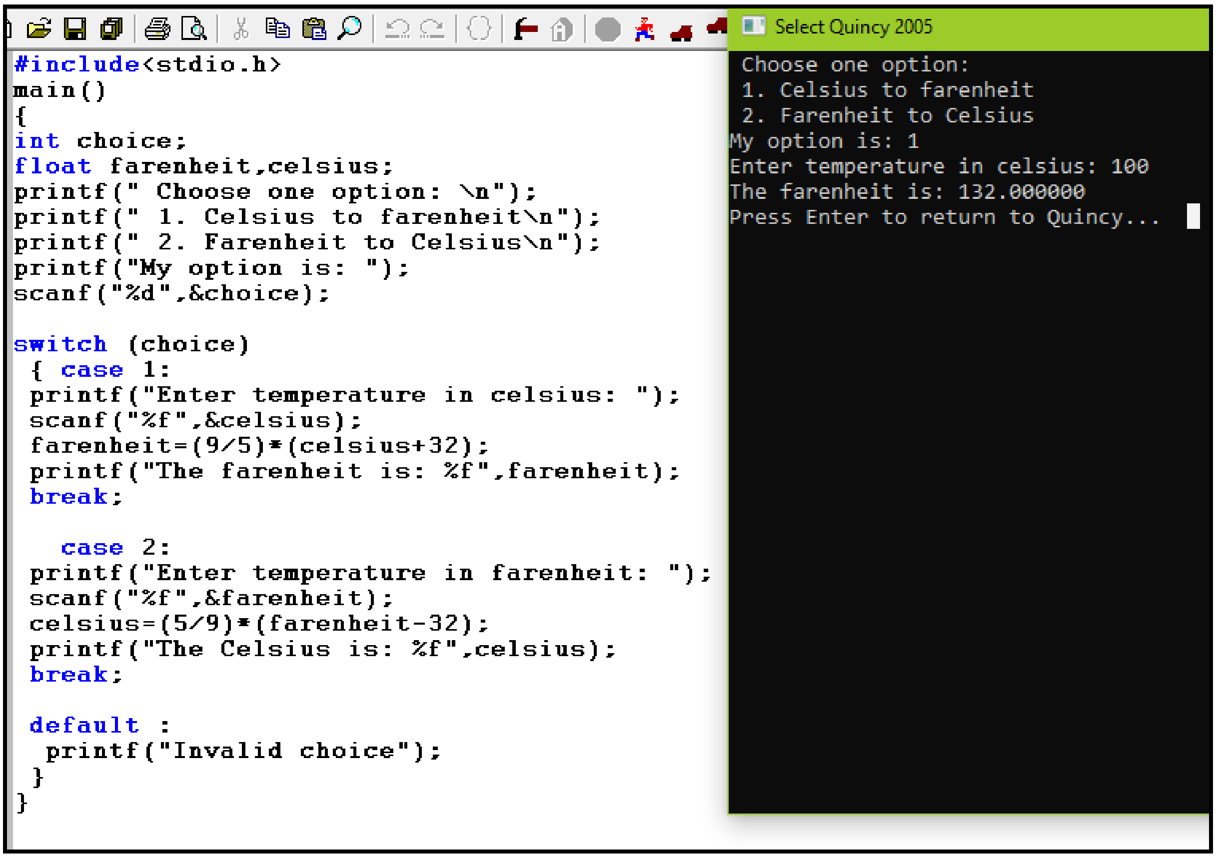
```c
#include<stdio.h>
main()
{
float P,R,T,I;
printf("enter the amount");
scanf("%f",& P);
printf("enter the rate");
scanf("%f",& R);
printf("enter the time");
scanf("%f",&T);
I=(P*R*T)/100;
Printf("the simple interest is %f", I);
}
```

5. Write a C program to print multiplication table of any number.

```c
#include<stdio.h>
main()
{
 int n,i;
 printf("Enter the table number: \n ");
 scanf("%d",&n);
  for(i=1;i<=12;++i)
  {
     printf("%d*%d=%d\n",n,i,n*i);
  }
return 0;
}
```

```
Quincy 2005
Enter the table number:
 4
4*1=4
4*2=8
4*3=12
4*4=16
4*5=20
4*6=24
4*7=28
4*8=32
4*9=36
4*10=40
4*11=44
4*12=48

Press Enter to return to Quincy...
```

6. Using switch condition statement, write a C program to convert degrees Fahrenheit to degrees Centigrade or degrees Centigrade to degrees Fahrenheit. The program should display two menu options on which a user can make a choice on what conversion he/she want to make. Hint: C = (5/9)(F – 32)

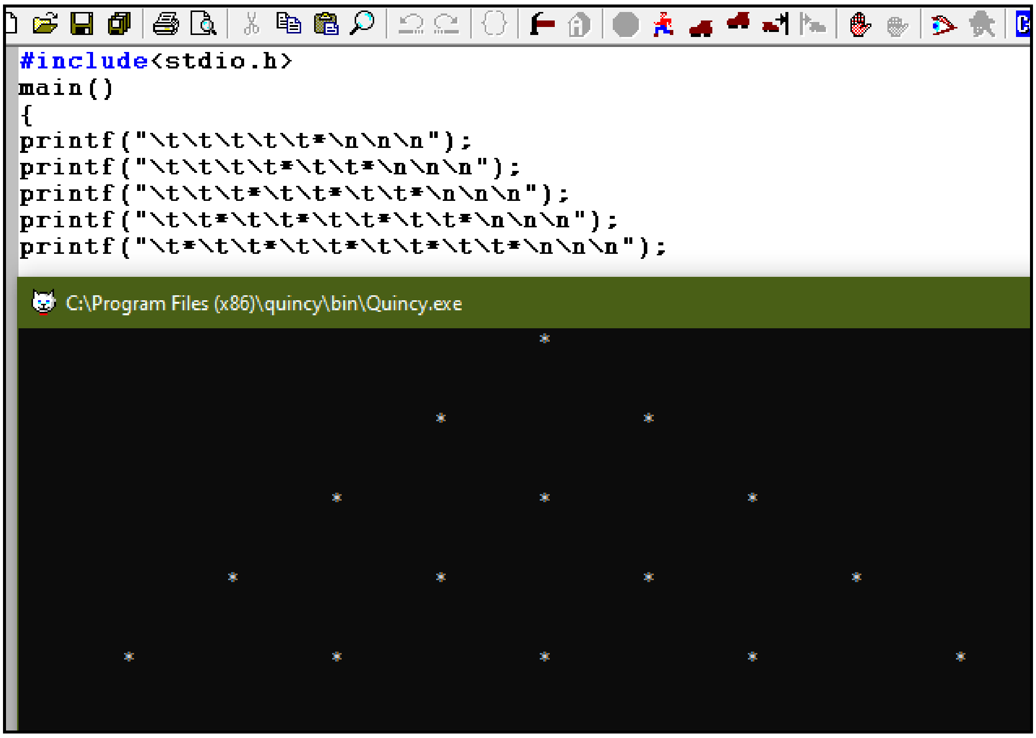


```
#include<stdio.h>
main()
{
int choice;
float farenheit,celsius;
printf(" Choose one option: \n");
printf(" 1. Celsius to farenheit\n");
printf(" 2. Farenheit to Celsius\n");
printf("My option is: ");
scanf("%d",&choice);

switch (choice)
 { case 1:
 printf("Enter temperature in celsius: ");
 scanf("%f",&celsius);
 farenheit=(9/5)*(celsius+32);
 printf("The farenheit is: %f",farenheit);
 break;

   case 2:
 printf("Enter temperature in farenheit: ");
 scanf("%f",&farenheit);
 celsius=(5/9)*(farenheit-32);
 printf("The Celsius is: %f",celsius);
 break;

 default :
  printf("Invalid choice");
 }
}
```

```
Select Quincy 2005
 Choose one option:
 1. Celsius to farenheit
 2. Farenheit to Celsius
My option is: 1
Enter temperature in celsius: 100
The farenheit is: 132.000000
Press Enter to return to Quincy...
```

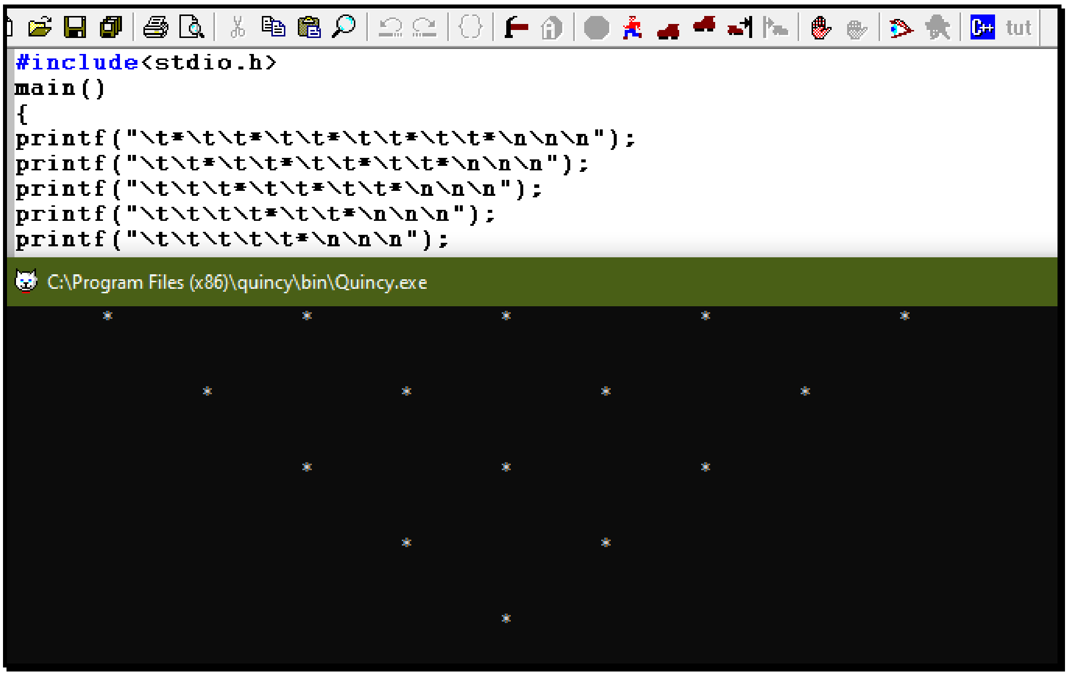
7. a. Write a C program to display the following output;

```
    *
   * *
  * * *
 * * * *
* * * *  *
```



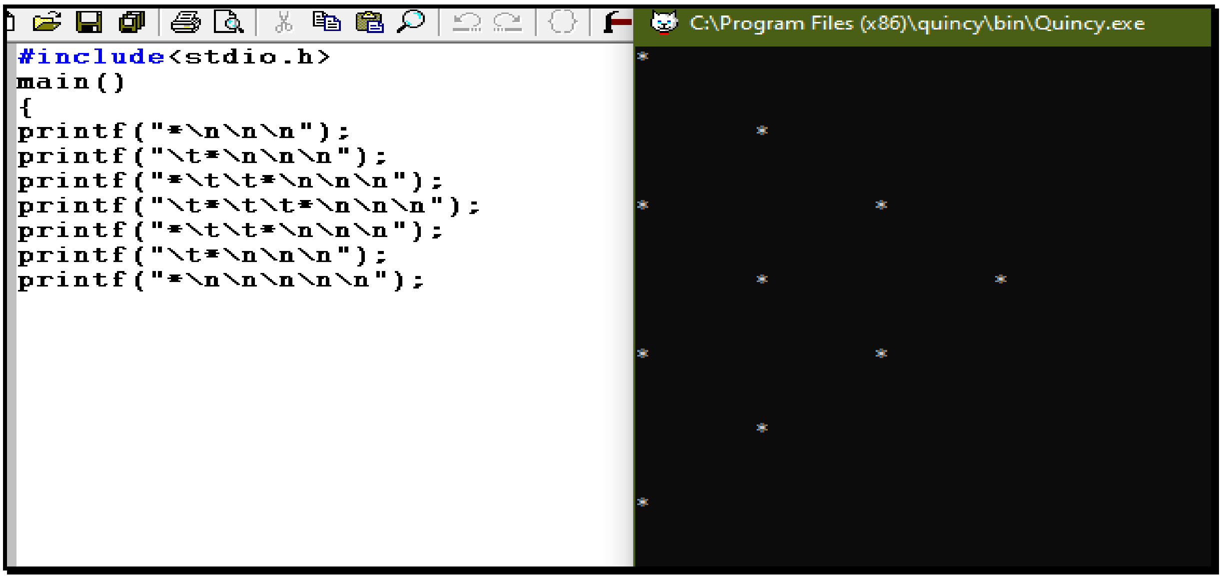
b. Write a C program to display the following output;

```
* * * *  *
 * * * *
  * * *
   * *
    *
```



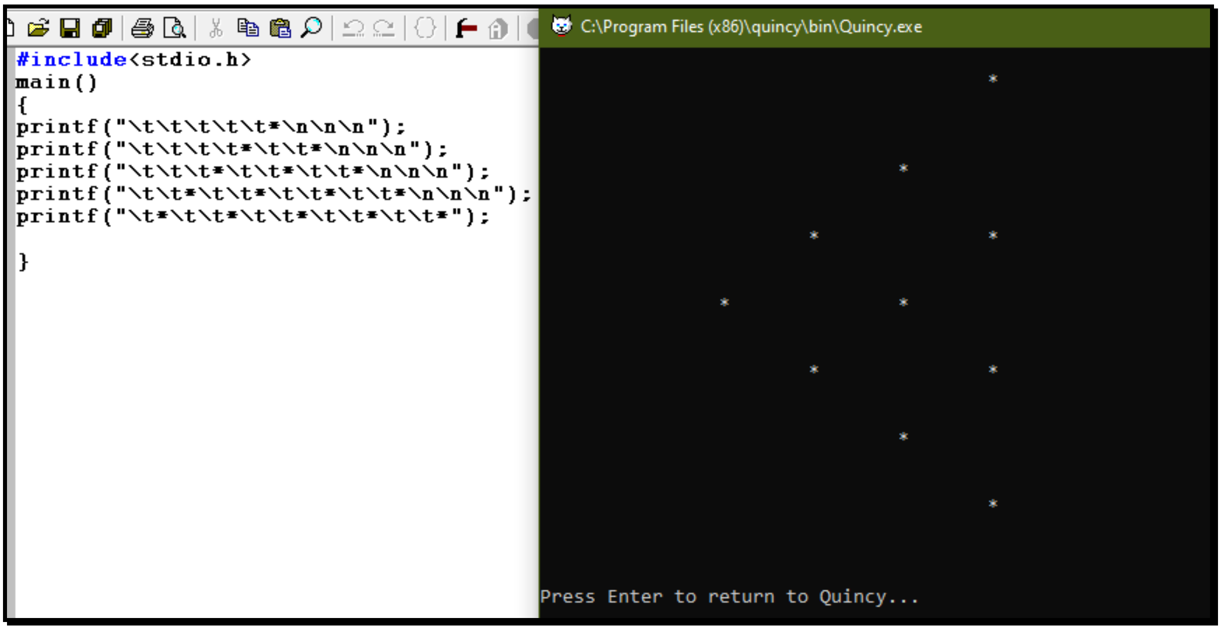
c. Write a C program to display the following output;

```
*

      *

*          *

      *          *

*          *

      *

*
```

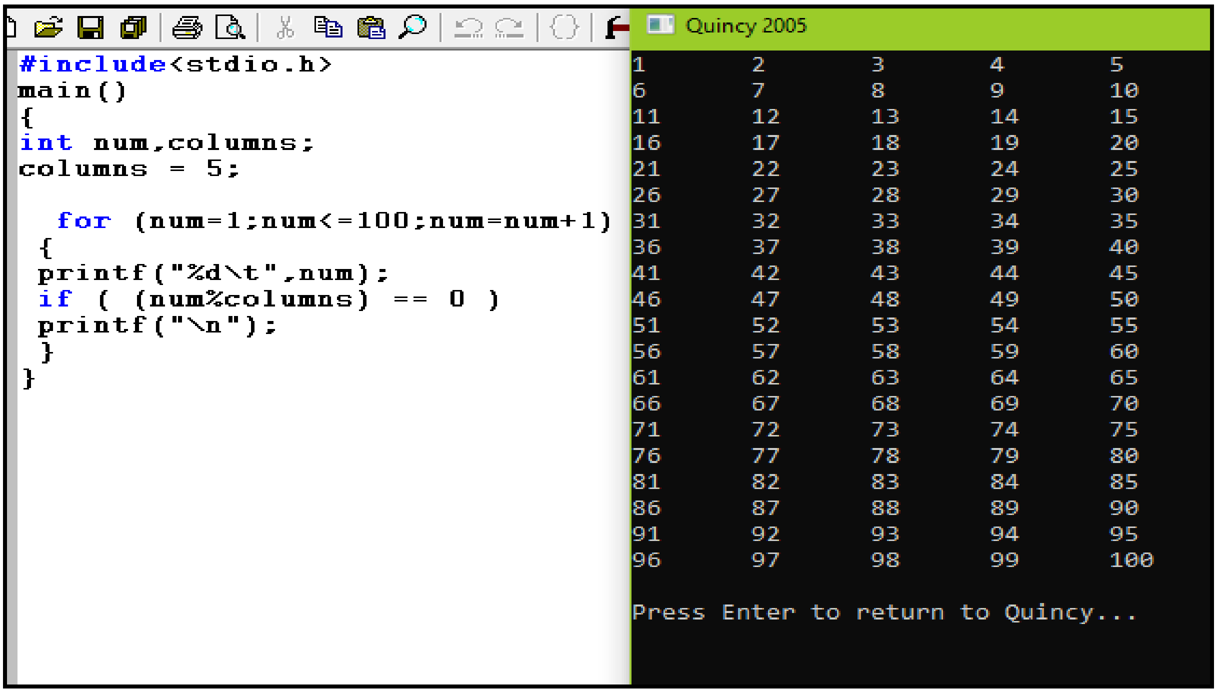


d. Write a C program to display the following output;

```
                  *

            *

      *           *

*           *

      *           *

            *

                  *
```



8. Write a C program that prints the numbers 1 to 100 using 5 columns. Have each number separated from the next by a tab.



9. Write a C program to display the following output

```
* * * *
 * * *
  * *
   *
```

```c
#include <stdio.h>
main()
{

 printf("     * * * *\n");
 printf("      * * *\n");
 printf("       * * \n");
 printf("        *\n");
 }
```

10. Write a C program to find the eligibility of admission for a professional course based on the following criteria: Eligibility Criteria: Grade in Maths and Physics should be minimum of C.

```c
#include < stdio.h >
main ( )
{
int grades;
printf ("Enter the Maths grades: \n");
scanf ("%d",& grades);
    {
      if ( grades >= 70 )
        printf ( "A\n\n" );
     else if ( grades >= 60 )
       printf ("B+\n\n");
```

```
        else if ( grades >= 50 )
          printf ("B\n\n");
        else if ( grades >= 40 )
          printf ("C\n\n");
        else if ( grades >= 0 )
         printf ("Admission is not eligible, since the grades haven't reached the
        target\n\n");
    }
    {
        printf ("Enter the Physics grades: \n");
        scanf ("%d",& grades);
         if ( grades >= 70 )
          printf ("A");
        else if ( grades >= 60 )
          printf ("B+");
        else if ( grades >= 50 )
          printf ("B");
        else if ( grades >= 40 )
          printf ("C");
        else if ( grades >= 0 )
      printf ("Admission is not eligible, since the grades haven't reached the target");
        }
    }
```

After running the program.

i.

```
Quincy 2005
Enter the Maths grades:
67
B+

Enter the Physics grades:
98
A
Press Enter to return to Quincy...
```

ii.

```
C:\Program Files (x86)\quincy\bin\Quincy.exe
Enter the Maths grades:
67
B+

Enter the Physics grades:
20
Admission is not eligible,since the grades haven't reached the target
Press Enter to return to Quincy...
```

iii.

```
Enter the Maths grades:
34
Admission is not eligible,since the grades haven't reached the target

Enter the Physics grades:
12
Admission is not eligible,since the grades haven't reached the target
Press Enter to return to Quincy...
```

11. Write a C program to read n numbers from keyboard and find their sum and Average.

```c
#include<stdio.h>
main()
{
int i,n,sum=0,average;
printf("\n   How many numbers to be executed ?: ");
scanf("%d",&n);

 for(i=1; i<=n; i++)
 {
 sum=sum+i;
 average=sum/n;
 }
printf("   The sum and average are: %d,%d \n",sum,average);

}
```

```
 How many numbers to be executed ?: 4
 The sum and average are: 10,2

Press Enter to return to Quincy...
```

```c
#include<stdio.h>
main()
{
int i,n,sum=0,average;
printf("\n   How many numbers to be executed ?: ");
scanf("%d",&n);

 for(i=1; i<=n; i++)
 {
 sum=sum+i;
 average=sum/n;
 }
printf("   The sum and average are: %d,%d \n",sum,average);

}
```

```
Quincy 2005
 How many numbers to be executed ?: 5
 The sum and average are: 15,3

Press Enter to return to Quincy...
```

Kuwen na maandalizi mema ya UE.